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Oefeningenreeks 4A nr. 36.13

$$\begin{aligned} & \int \frac{5dx}{x^2 + 5} \\ &= 5 \int \frac{dx}{x^2 + 5} \\ &= 5 \int \frac{dx}{5 \left(\frac{x^2}{5} + 1 \right)} \\ &= 5 \cdot \frac{1}{5} \int \frac{dx}{\frac{x^2}{5} + 1} \\ &= \int \frac{dx}{\left(\frac{x}{\sqrt{5}} \right)^2 + 1} \\ &t = \frac{x}{\sqrt{5}} \Rightarrow dt = \frac{1}{\sqrt{5}} dx \Rightarrow \sqrt{5} dt = dx \\ &= \sqrt{5} \int \frac{dt}{t^2 + 1} \\ &= \sqrt{5} \operatorname{Bgtan}(t) + C \\ &= \sqrt{5} \operatorname{Bgtan} \left(\frac{x}{\sqrt{5}} \right) + C \end{aligned}$$

References